THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 08

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 17 units	$\sqrt{(10-(2))^2+(14-(-1))^2}=\sqrt{64+225}=17$
2	**(a)** False; they are different because $(x,y)\neq(y,x)$ when $x\neq y$	order of coordinates matters
3	**(c)** on the x-axis	$y=0$, so the point lies on the x-axis
4	**(a)** 1	highest power of the variable is 1
5	**(b)** does not lie on the line	RHS at $x=2$ is 4, not equal to 5
6	**(a)** False; $\sqrt{2}\times\sqrt{2}=2$ is rational	product of irrationals can be rational
7	**(a)** irrational	non-terminating and non-repeating
8	**(d)** 3	number of places $=\max(a,b)$ in 2^a5^b
9	**(d)** $4n^2-49$	$(a+b)(a-b)=a^2-b^2$
10	**(b)** 2	$(a+b+c)^2=a^2+b^2+c^2+2ab+2bc+2ca$
11	**(a)** 50°	angle at the centre is double the angle on the circle
12	**(a)** 15 cm	$r=\sqrt{9^2+12^2}=15$
13	**(d)** 24 cm	$d=\sqrt{25^2-7^2}=24$
14	**(c)** πr	half of $2\pi r$
15	**(c)** 84 cm^2	$s=21$, area $=\sqrt{21\cdot8\cdot7\cdot6}=84$
16	**(b)** 88 cm	$2\pi r=2\times\frac{22}{7}\times14=88$
17	**(b)** 4	multiply the number of outcomes of each experiment
18	**(b)** theoretical probability	Law of Large Numbers
19	**(c)** 13	$T_4=7$, $T_5=4+2+7=13$
20	**(c)** 39	$a+(n-1)d=-5+11\times(4)=39$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Midpoint $=\left(\frac{0+(-8)}{2},\frac{0+(10)}{2}\right)=(-\mathbf{4},\mathbf{5})$	1+1
22	$2n-1$. $2n-1=47\Rightarrow n=24$.	1+1
23	12 has a prime factor other than 2 and 5 ⇒ non-terminating repeating	1+1
24	$(3x+4y)^2$	2
25	$5x+30=180\Rightarrow x=30$; $\angle A=90^\circ$.	1+1
26	$\frac{60}{360}\times2\times\frac{22}{7}\times3.5=\frac{11}{3}\approx3.67$ cm.	1+1
27	36 equally likely outcomes; sum 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) ⇒ $\frac{6}{36}=\frac{1}{6}$.	1+1
28	Yes — every term is -1 times the previous one: $r=-1$, $t_n=(-1)^{n-1}$.	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Sides: 4, 3 and $\sqrt{16+9}=5$. Perimeter $=12$ units.	2+1
30	Width $=10-x$; area $=x(10-x)=10x-x^2$ (a quadratic polynomial). At $x=6$: $60-36=24$ cm^2.	2+1
31	$\frac{a+b}{2}-a=\frac{b-a}{2}>0$ and $b-\frac{a+b}{2}=\frac{b-a}{2}>0$. So $a<\frac{a+b}{2}<b$; it is rational as sums and quotients of rationals are rational.	1+1+1
32	$6\times2=12$; split $7x=3x+4x$: $6x^2+3x+4x+2=3x(2x+1)+2(2x+1)=(2x+1)(3x+2)$.	1+1+1
33	$r=14$ m (1); area $=\frac{22}{7}\times196=616$ m^2 (1); cost $=88\times12=$ ₹1056 (1).	1+1+1
34	(i) 2, 3, 5, 7 ⇒ $\frac{4}{8}=\frac{1}{2}$ (ii) 6, 7, 8 ⇒ $\frac{3}{8}$.	1+1+1
35	B lies on the circle outside arc ADC, so the angle at the centre is double: $2\times110^\circ=220^\circ$.	2+1

	OR: Let $AB = CD$ with perpendiculars OM, ON from O. Then $AM = \frac{1}{2}AB = \frac{1}{2}CD = CN$ (1). In right triangles OMA, ONC: $OA = OC$, $AM = CN$ ⇒ congruent (RHS) (1), so $OM = ON$ (1).	1+1+1
36	$1 + 3 + 5 + 7 + 9 = 25$ ✓ (1). For the AP $1, 3, \ldots, (2n-1)$: $S = \frac{n}{2}(1 + (2n-1)) = n^2$ (2).	1+2
	OR: $5n - 2 = 308 \Rightarrow n = 62$: yes (1+). $5n - 2 = 471 \Rightarrow n = 94.6$, not a natural number ⇒ not a term (2).	1+2

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) -12 and 12. (ii) $-850 + 1200 - 450 = -100$ ⇒ debt of ₹100. (iii) Removing 4 debts of ₹3 each makes one ₹12 richer.	2+2+1
	OR: (i) $\frac{126}{10} = \frac{63}{5}$. (ii) $\frac{120}{10000} = \frac{3}{250}$. (iii) $\frac{3052}{1000} = \frac{763}{250}$. (iv) $\frac{23}{99}$. (v) $x = 2.0\overline{5}$: $10x = 20.\overline{5}$, $100x = 205.\overline{5}$, $90x = 185 \Rightarrow x = \frac{37}{18}$.	1+1+1+1+1
38	Let $OM = ON$ with $OM \perp AB, ON \perp CD$. Right triangles OMA, ONC: $OA = OC$, $OM = ON$ ⇒ congruent (RHS) ⇒ $AM = CN$ (2). Halves equal ⇒ $AB = CD$ (1). Length $= 2\sqrt{169 - 144} = 10$ cm (2).	2+1+2
	OR: (i) The arc on the other side of the diameter subtends $180°$ at the centre, so the inscribed angle is $90°$ (2). (ii) Centre: $2 \times 62° = 124°$ (1). ACBD is cyclic so $\angle ADB = 180° - 62° = 118°$ (2).	2+1+2
39	(i) If the diameters are $d_1, \ldots, d_n$ then $d_1 + \cdots + d_n = PQ$; length $= \pi(d_1 + \cdots + d_n)/2 = \pi PQ/2$, the same as the big semicircle (3). (ii) $\frac{22}{7} \times \frac{28}{2} = 44$ cm; also $\frac{22}{7} \times \frac{7+7+14}{2} = 44$ cm ✓ (2).	3+2
	OR: (i) Construct a square with the same area as the rectangle (1). (ii) Area $= 100$ cm^2; side $= 10$ cm (2). (iii) Square perimeter $= 40$ cm; rectangle perimeter $= 58$ cm — the square has the smaller perimeter for the same area (2).	1+2+2
40	(i) $\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = \frac{n(2n)}{2} = n^2$ (2). (ii) $55 + 45 = 100$ ✓ (1). (iii) $t_{17} = 153$, $t_{80} = 3240$ (2).	2+1+2
	OR: (a) AP, $d = 4$, $t_n = 4n + 3$ (1). (b) GP, $r = 3$, $t_n = 2 \cdot 3^{n-1}$ (1). (c) Neither (differences 3,5,7; ratios 4, 2.25) but $t_n = n^2$ (1). (d) GP, $r = \frac{1}{2}$, $t_n = 100\left(\frac{1}{2}\right)^{n-1}$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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